Project: Azure Capacity Reservation Management Engine (ACRME)
Classification: Principal Cloud Architect - Architecture Governance
Version: 2.4
Date: 7 September 2026
Status: Accepted - supersedes ADR-003 v1.2 fixed-ratio DR model; reconciled to Baseline v2.4
Part of: ACRME Architecture Decision Records - aligned to Capacity & Quota Management Requirements Baseline v2.4.

> **About ADRs.** An Architecture Decision Record captures a significant architectural decision, the context that forced it, the options considered, the choice made, and its consequences. This ADR updates disaster-recovery capacity management to the lean bootstrap and distributed DR model. Evidence tags: `[Documented]`, `[Decided]`, `[Derived]`, `[Assumed]`.
>
> **v2.4 reconciliation note.** The DR capacity model (max-not-sum sizing, `SourceDestinationDRIndex`, standby activation waves, CVAL earmark / no-double-count) is **unchanged** by Baseline v2.4. v2.4 adds the **even per-zone distribution target and rebalancing action (PLC-011, Appendix A.9)** and the **regional + per-AZ CRG structure (CAP-023)**; because DR sizing here is computed **per zone**, this ADR now cross-references PLC-011/A.9 at the sizing formula so per-zone DR floors and per-AZ CRG sizing stay balanced (see DR Sizing Formula below).

ADR-003 - Capacity Management during Disaster Recovery (DR)

Status: Accepted
Date: 27 August 2026
Deciders: Principal Cloud Architect, DR Owner, Platform Engineering, Security, FinOps
Related requirements: ENV-003..ENV-006, DR-001..DR-019, PLC-010, DAT-002, DAT-003, OBS-001..OBS-004, OPS-001
Related POCs/decisions: POC-001, POC-006, POC-007, POC-011, DEC-001, DEC-002, DEP-001

Context

The v1 DR model assumed a fixed 30-40% reserve relative to production. Requirements v2.1 explicitly rejects that as the default because idle DR reservations at platform scale are too expensive and do not reflect the single-region-failure assumption. `[Decided]`

ACRME now needs a DR model that:

- starts with configurable bootstrap capacity, not a percentage clone of production;
- reconciles reserved capacity dynamically against allocated demand and policy buffers;
- uses CVAL/NonProd as a potential DR capacity source only under authorised runbooks;
- supports reciprocal multi-source DR hosting, where any region may simultaneously host Prod, CVAL, and DR standby for different customers;
- sizes each DR destination using max-over-non-concurrent-source demand;
- activates the correct standby set by source-region failure; and
- preserves auditability and failback reversibility. `[Derived]`

Decision

Adopt a **lean bootstrap plus distributed DR capacity architecture**:

1. **DR starts lean.** DR holds a configurable bootstrap target by workload, product, region, zone, SKU/family, and subscription model. It must not default to a fixed 30-40% copy of production. Zero bootstrap is allowed only through explicit approved policy. `[Decided]`

2. **Destination sizing uses max-not-sum.** For a destination region, standby capacity is sized to the largest non-concurrent protected source workload portion, not the sum of all sources. `[Decided]`

3. **SUM remains a conservative override.** If a customer contract, geography, or risk decision requires simultaneous source-region failure coverage, policy may switch that scope to SUM sizing. `[Decided]`

4. **Every region can have three concurrent roles.** A region may host its own production, CVAL for customers whose production is elsewhere, and standby DR for customers from multiple source regions. `[Decided]`

5. **CVAL is a DR capacity source, not free capacity.** CVAL capacity can be shut down, disassociated, or reassigned only on an authorised DR declaration and per approved priority/runbook. `[Decided]`

6. **DR activation is per customer and priority wave.** A declared region failure changes engine mode, then ACRME activates the affected source region's mapped standby instances from associated/inactive state to allocated/active state in approved business-priority waves. `[Decided]`

7. **Failback is reversible and audited.** Every activation records customer state, source/destination, capacity acquisition path, approvals, and rollback/failback checkpoint. `[Decided]`

Distributed DR Reference Model

The authoritative mapping is `SourceDestinationDRIndex`:

Field	Purpose
`source_region`	Failed or protected production source.
`destination_region`	Region holding standby DR capacity.
`customer_or_realm_id`	Customer/realm whose standby set is mapped.
`seed_id`	Link back to `CustomerSeedRecord` from ADR-001.
`standby_instance_set`	VM/VMSS/node-pool/application set eligible for activation.
`sku_family`, `sku`, `zone`, `quantity`	Capacity/quota dimensions.
`activation_state`	`standby`, `activation_pending`, `active`, `failback_pending`, `returned`.
`priority_wave`	Business priority for activation order.
`policy_version`, `last_updated`	Replay, freshness, and audit fields.

The index must be queryable in both directions:

- source -> destinations, to know where a failed region's customers activate;
- destination -> sources, to compute max-source coverage and dashboard exposure. `[Decided]`

DR Sizing Formula

For each destination, SKU, zone, and policy scope:

```
Destination_DR_Requirement(d)
  = MAX over non-concurrent source regions s protected by d (
      Workload_Portion(s -> d)
    )
```

With vCPU expansion:

```
DR_Floor_vCPU(d, sku, zone)
  = Destination_DR_Requirement(d, sku, zone) * vCPU_Per_Instance(sku)
```

The old formula `prod_vm_count * dr_ratio_max` is superseded. It may appear only in legacy examples, never as the default production sizing rule. `[Decided]`

Zone-distribution interaction (PLC-011, Appendix A.9) — cross-reference. Because the DR floor above is computed **per zone**, the per-zone `Workload_Portion(s → d)` values depend on how each source workload is spread across its availability zones. Under **PLC-011** placement targets an **even ≈ `1/zone_count` spread** of a workload's VMs across zones (≈ 33% each in a three-zone region), and raises a **rebalancing action** when a deployment or growth event skews distribution beyond the configured tolerance (C-13); see **Appendix A.9** of the baseline for the skew formula. Two consequences for DR sizing: (a) an **even source distribution** keeps the per-zone `Destination_DR_Requirement(d, sku, zone)` balanced, so no single destination zone carries a disproportionate max-not-sum floor; and (b) the resulting per-zone DR reservations are held in the destination's **per-AZ CRGs (CAP-023)** — the per-AZ CRG for each zone sizes to that zone's DR floor, and the regional CRG holds only SKUs without zonal reservation support. A skewed source distribution therefore inflates one destination zone's DR floor; the PLC-011 rebalancing action is the corrective control, and DR sizing consumes the (rebalanced) per-zone portions rather than re-deriving distribution. `[Derived]`

DR Activation Sequence

```
flowchart TD
    Declare[Authorised DR declaration] --> Mode[Set DR_EVENT_ACTIVE]
    Mode --> Index[Read SourceDestinationDRIndex]
    Index --> Waves[Order customers by priority wave]
    Waves --> Bootstrap[Use bootstrap capacity]
    Bootstrap --> Quota[Use available destination quota]
    Quota --> CVAL[Release approved CVAL earmark]
    CVAL --> Share[Use approved shared reservation]
    Share --> Pool[Allocate pooled quota]
    Pool --> AzureReq[Submit Azure request if still deficient]
    AzureReq --> Activate[Transition standby associated to allocated active]
    Activate --> Audit[Record activation and failback checkpoint]
```

Capacity acquisition order:

1. existing bootstrap capacity;
2. available destination quota/capacity;
3. approved CVAL release;
4. approved capacity sharing;
5. pooled quota allocation;
6. Azure quota/capacity request with exposed deficit if Azure cannot supply it. `[Decided]`

CVAL Earmark and No-Double-Count Rule

When CVAL and DR co-locate, ACRME must create a `CVALEarmarkRecord` that identifies which CVAL capacity is releasable for a customer's DR activation. Earmarked CVAL capacity counts toward DR headroom, not live CVAL headroom. It must never be credited as both live CVAL capacity and available DR capacity. `[Decided]`

`CVALEarmarkRecord` contains customer/realm, CVAL region, DR destination, SKU/zone/quantity, release action, approval policy, priority wave, expiry/review status, and audit references. `[Decided]`

Engine State Machine

`engine_mode` is a persisted state machine with conditional writes and operator-gated transitions:

State	Meaning	Permitted transitions
`STEADY_STATE`	Normal inventory, reconciliation, placement, and approved growth.	`DR_DECLARATION_PENDING`
`DR_DECLARATION_PENDING`	DR requested and awaiting approval/validation.	`DR_EVENT_ACTIVE`, `STEADY_STATE`
`DR_EVENT_ACTIVE`	Source-specific DR activation and emergency operations permitted.	`FAILBACK_PENDING`, `INCIDENT_HOLD`
`FAILBACK_PENDING`	Return/reversal is being validated and executed.	`STEADY_STATE`, `DR_EVENT_ACTIVE`, `INCIDENT_HOLD`
`INCIDENT_HOLD`	Safe degraded state after conflict or critical failure.	Recovery-approved transition only

Entering `DR_EVENT_ACTIVE` does not automatically authorise service-impacting CVAL action, Tier 2, or Tier 3. Each action still requires its own policy gate. `[Decided]`

Relationship to Emergency Tiers

The previous three-tier emergency transfer model remains as an implementation pattern, but v2.1 scopes it behind bootstrap-first activation:

Tier	v2.1 treatment
Tier 1 - direct expansion	Permitted after DR declaration when quota/capacity/sharing preconditions are fresh.
Tier 2 - quota-neutral transfer	Approval-gated until quota group release and consumer quota behavior are proven.
Tier 3 - destructive VM association changes	Blocked in Phase 1; VMSS entries are rejected.

Observability and Runbooks

ACRME dashboards and alerts must show:

- DR bootstrap below target;
- destination coverage below max protected source;
- source <-> destination mapping;
- activation state by customer and priority wave;
- CVAL earmarks and releasable capacity;
- quota/capacity deficits after each acquisition stage;

- failed/stale activation state;
- failback pending age. `[Decided]`

Runbooks must cover DR declaration, standby activation, CVAL release/shutdown/disassociation, quota allocation to DR, sharing activation, manual emergency override, regional recovery, and failback. `[Decided]`

Consequences

Positive

- Removes the high-cost fixed 30-40% standby reserve as the default. `[Decided]`
- Aligns standby sizing with the single-region-failure assumption. `[Derived]`
- Supports reciprocal multi-source hosting without summing unrelated non-concurrent sources. `[Decided]`
- Makes activation deterministic from the DR index and seed records. `[Decided]`

Negative / trade-offs

- Max-not-sum under-protects two simultaneous source-region failures unless SUM override is configured. `[Derived]`
- Capacity Reservation Sharing remains a preview/feature-maturity dependency for parts of the cost model. `[Assumed]`
- Bootstrap sizing requires workload-specific POC evidence. `[Assumed]`
- CVAL release can be service-impacting and therefore requires explicit authorisation. `[Decided]`

Alternatives Considered

Alternative	Disposition
Fixed 30-40% production copy	Rejected as default because of idle cost and v2.1 pivot. `[Decided]`
Sum all sources per destination	Rejected as default; too expensive under single-region-failure assumption. `[Decided]`
Zero DR bootstrap by default	Rejected; zero is allowed only by explicit approved policy. `[Decided]`
Treat CVAL free capacity as always available	Rejected; requires earmark, authorisation, and no-double-count controls. `[Decided]`
Activate all destination standby capacity during any event	Rejected; activation is source-specific via the DR index. `[Decided]`

Appendix - ADR Summary

ADR	Requirements Applied	Key Open Items
ADR-003 Capacity Management during DR	ENV-003..006, DR-001..019, PLC-010, DAT-002, OBS-004	POC-006 topology, POC-007 bootstrap sizing, POC-011 max-not-sum safety, DEC-001 Middle East DR, DEC-002 fail-back duration, DEP-001 sharing maturity

Appendix - Status Legend

Status	Meaning
Proposed	Under discussion; not yet ratified
Accepted	Ratified and in force
Deprecated	No longer recommended but not yet replaced
Superseded	Replaced by a later ADR

Appendix - Evidence Tag Taxonomy

Tag	Meaning
`[Documented]`	Traceable to Azure platform behaviour or documentation
`[Decided]`	An explicit ACRME design choice recorded in this ADR set
`[Derived]`	A logical consequence of a documented constraint or decision
`[Assumed]`	Architectural judgement pending proof-of-concept validation

Related ADRs

- **ADR-001 - Region Selection and Customer Placement** (`acrme_adr_001_region_selection.md`)
- **ADR-002 - Quota and Capacity Management** (`acrme_adr_002_quota_and_capacity_management.md`)
- **ADR-004 - Forecast, Reconciliation, and Increase of Capacity and Quota** (`acrme_adr_004_forecast_and_increase_of_capacity_and_quota.md`)

Document Status: Accepted
Next Review: After POC-006, POC-007, POC-011, DEC-001, DEC-002, and Capacity Reservation Sharing maturity review.